Depth Anything V2-Based Proximity Warning System

Introduction

This task implements a video-based proximity warning system using Depth Anything V2 Metric Small (ViT-S). The objective is to estimate scene depth and identify potentially nearby objects within a selected region of interest (ROI).

Methodology

The experiment was conducted on a driving video from the **Roadscapes dataset**[1], specifically *Day 11*. The original video was trimmed to include the interval from 6 to 50 seconds, resulting in a 44-second video containing 1,320 frames. The video has a resolution of 1280 × 720 pixels and a frame rate of 30 FPS.

For each frame, the model generates a depth map. The ROI covers 80% of the image width, from 10% to 90%, and 70% of its height, from 25% to 95%. The 10th percentile (P10) of valid depth values within this region is used as an indicator of nearby scene content.

A warning threshold of 8 meters was selected based on several preliminary tests. The warning is activated after three consecutive frames below the threshold and cleared after three consecutive frames above it. When the warning is active, the output video displays **"WARNING! OBJECT TOO CLOSE"**. The system also records depth statistics and warning status in a CSV file.

Results

The system generated a proximity-warning video, a depth visualization video, and a CSV file containing results for all 1,320 frames. The average processing speed was approximately 4.90 FPS, below the input rate of 30 FPS; therefore, the current implementation does not achieve real-time processing.

According to the CSV results:

- **Warning activations:** 8
- **Warning clearances:** 7
- **Frames with an active warning:** 759 (57.5%)
- **Total warning-active time:** approximately 25.3 seconds
- **P10 range across all frames:** approximately 3.08–9.33 meters
- **Mean P10:** approximately 7.41 meters

The warning was first activated at 0.07 seconds and remained active at the end of the video. Subsequent activations occurred at approximately 1.53, 3.30, 3.77, 9.70, 13.80, 33.73, and 38.90

[1] https://huggingface.co/datasets/vijpandaturtle/roadscapes-video

seconds. The warning cleared at approximately 0.73, 2.10, 3.67, 9.07, 12.47, 23.73, and 34.33 seconds.

These results demonstrate that the system can generate temporally stabilized warnings based on depth estimates within the selected ROI. However, the proportion of warning-active frames also indicates that the chosen threshold produces frequent warnings in this particular video.

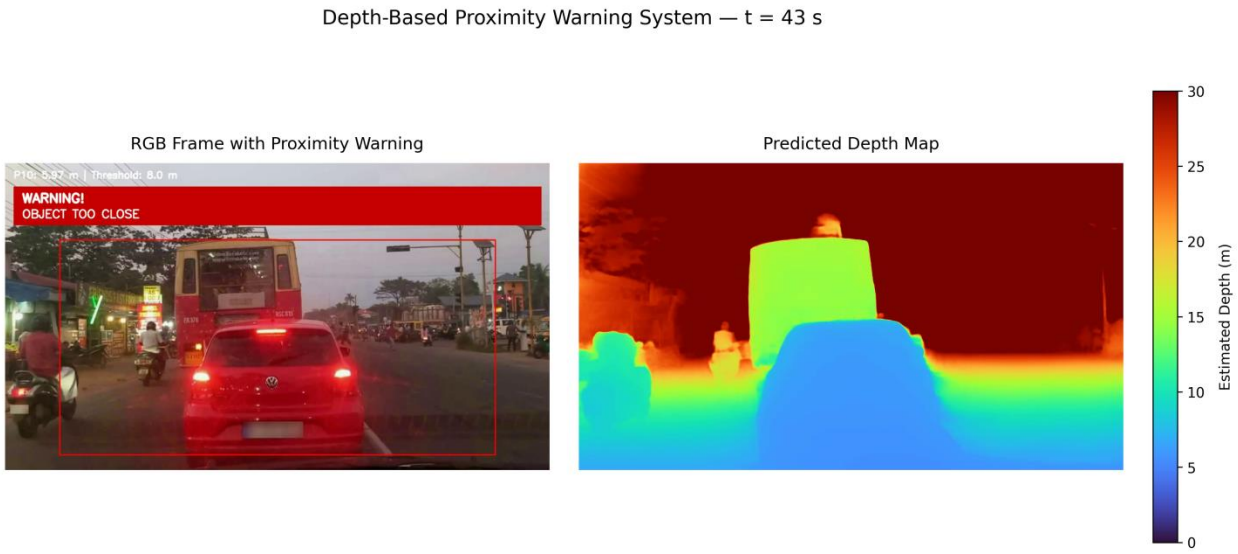


Figure 1. Proximity warning system output at (t = 43) s. Left: RGB frame with a warned close object. Right: corresponding Depth Anything V2 predicted depth map, color-coded by estimated distance in meters.

Limitations

The system uses depth statistics rather than explicit object detection or tracking. Consequently, the P10 value may be affected by different objects or scene regions within the ROI. In addition, the model's predictions may be affected by differences between its training domain and the driving footage. The 8-meter threshold is an experimental operating parameter, not a validated vehicle-safety distance. The processing speed also remains below real-time requirements.

Conclusion

This experiment demonstrates the integration of Depth Anything V2 into an ROI-based video proximity warning pipeline. The system generates depth maps, evaluates the lower end of the ROI depth distribution, and uses a three-frame rule to stabilize warning transitions. Future work could investigate object-specific depth estimation, validation against ground-truth distances, threshold calibration, and inference optimization to improve reliability and processing speed.